
MONSTRUMS GAMING · QUALITY ASSURANCE

TAP TRADE · QA SPEC

Test specification and acceptance criteria for Tap Trade — every rule, exact expected value, toast string, and money movement, plus the list of intentional behaviors that must not be filed as bugs. Companion to the internal GDD.

DOCUMENT	QA Test Specification — internal
GAME ID	2_6_tap_trade (math) · 2-6-tap-trade (web)
VERSION	1.0 · 2026-07-22
SURFACES	Web app (primary) · math-sdk frontend_demo (reference)
CASE PREFIX	TT-* · 74 cases across 11 suites

Companion: docs/Tap_Trade_GDD.pdf — design & architecture rationale

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1 Environments & Launch Modes

Web app · dev	<code>pnpm run dev --filter=2-6-tap-trade</code> → <code>http://localhost:3026/</code> — offline mock RGS auto-installs (dev builds only). Start balance \$2,000.
Web app · prod build	<code>pnpm run build --filter=2-6-tap-trade</code> then <code>pnpm run preview</code> — paramless launch must BLOCK (see TT-ERR-05); <code>?demo=1</code> unlocks play-money.
Web app · LIVE	Launch with <code>?rgs_url=<host>&sessionID=<id>&currency=USD&lang=en</code> . Badge shows LIVE .
Locale testing	Append <code>&currency=EUR&lang=de</code> (or JPY, or a crypto code) — see TT-BET-05...07.
Reference demo	<code>math-sdk/games/2_6_tap_trade/frontend_demo</code> : <code>npm run dev</code> . When behavior differs, the demo is the spec — file the web app.
Automated	<code>pnpm run test --filter=2-6-tap-trade</code> (22 vitest) · <code>pnpm run storybook</code> → :6026 (CellCallBook win/lose/20×/100× stories) · <code>pnpm run lint / build</code> .
Debug handles	DEV BUILDS ONLY: <code>window.__tt</code> (<code>.simTime()</code> , <code>.cfg</code> , <code>.tapState.chips</code>) and <code>window.__ttHistory</code> (entries). Their <i>absence</i> in a production build is itself a test (TT-ERR-07).
Reset state	<code>localStorage.clear()</code> removes <code>taptrade.*</code> keys (betSize, theme, muted, seenHint, speed, tapConfirm). Balance/history reset on reload (in-memory).

Pointer profile matters: tap-to-confirm defaults ON for coarse pointers (touch) and OFF for mouse. When a single click doesn't place a chip, check Settings → "Tap to confirm" before filing — the first tap only previews (TT-SET-05).

2 Smoke Suite (must pass on every build)

ID	STEPS	EXPECTED
TT-SMK-01	Load dev app, no params	Splash (loader → sting), then chart running, balance pill \$2,000.00 + LOCAL badge, hint visible on first visit, zero console errors (favicon 404 excepted)
TT-SMK-02	Tap a valid future cell (confirm twice if tap-confirm on)	Chip appears pulsing; balance drops by exactly the bet size, once
TT-SMK-03	Wait for the chip's column	Chip goes hot (countdown bar), line visibly hits (win) or misses (loss); win pays at the reveal moment, loss changes nothing
TT-SMK-04	After resolution	Session strip (top-right pill) shows P/L, "nothing in play", and a result pip; strip stays visible from now on
TT-SMK-05	Open history (clock button)	One row with a JPEG thumbnail, amount, multiplier · stake, timestamp
TT-SMK-06	Click the row	Modal: animated replay into the cell + landing shot + caption; Replay re-runs; Escape closes
TT-SMK-07	Open settings (gear)	Three rows: Game speed 1x/1.5x/2x · Sound switch · Tap to confirm switch
TT-SMK-08	Switch theme	Canvas AND all chrome re-skin instantly; toast "Theme: <name>"; persists across reload
TT-SMK-09	Wheel-zoom out fully (0.5x)	Grid/cells shrink; every placed chip still shows a readable (single-line) stake label; odds labels hide before colliding
TT-SMK-10	<code>pnpm run test/lint/build</code>	22 tests pass, lint clean, build green

3 Placement Rules (TT-PLC)

Every rejection shows a warn toast; cell-rule rejections also flash the cell red. Rejections must cost nothing (no balance change, no chip).

ID	ACTION	EXPECTED TOAST	RED FLASH
TT-PLC-01	Tap a cell in the shaded dead zone (< 2.5 s ahead)	"Too close to the line"	yes
TT-PLC-02	Tap a cell that already has a live chip	"Chip already there"	yes

ID	ACTION	EXPECTED TOAST	RED FLASH
TT-PLC-03	Tap a 3rd distinct cell in a column with 2 live chips	"Column full — max 2 chips"	yes
TT-PLC-04	Set bet > displayed balance, tap	"Insufficient balance"	no (toast only)
TT-PLC-05	Two chips in one column on distinct cells	Both accepted; both resolve correctly in the same pass	—
TT-PLC-06	Chips in different columns (3+)	All accepted — never rejected for count across columns; rounds settle serially (watch sequential <code>/wallet/play</code>)	—
TT-PLC-07	Resolved (won/lost) chips in a column	Do NOT count toward the 2-chip cap	—
TT-PLC-08	Queued chips + balance	Each queued chip reserves its stake — a 3rd chip that exceeds the remaining display balance is rejected (TT-PLC-04)	—

4 Wallet & Ledger (TT-LED) — the critical suite

The displayed balance is an optimistic ledger. The **only** legal movements:

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place chip    → balance - stake      (immediately, exactly once)
win REVEAL    → balance + payout     (at the line-fill moment, not at settlement)
loss reveal   → no change            (the stake already left at placement)
quiescence    → balance := server RGS balance (silent reconcile; normally a no-op)

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ID	SCENARIO	EXPECTED
TT-LED-01	Place one chip, let it LOSE	One deduction at placement; NO second deduction at the explode animation; "Missed" float is a marker, not a debit
TT-LED-02	Place one chip, let it WIN	Credit lands exactly when the line fills the cell — throttle the network (LIVE/mock) to prove the gap between settlement and reveal shows no early credit
TT-LED-03	3 chips across 2 columns, mixed outcomes	3 immediate deductions; credits only at each reveal; after the last reveal the pill EQUALS the server balance to the cent (LIVE: last <code>/wallet/*</code> response <code>balance.amount ÷ 1e6</code>)
TT-LED-04	Out-of-order settlement (chips in far column settle while near column reveals)	The pill NEVER jumps to a stale per-round snapshot — regression for the "\$1,010 vs \$1,005" drift class
TT-LED-05	Force <code>/wallet/play</code> failure (devtools block) after placing	Chip removed from the board, stake refunded on the pill, error modal/toast shown; LATER chips unaffected
TT-LED-06	Session strip math	$P/L = \Sigma(\text{payouts}) - \Sigma(\text{stakes})$; "\$X in play · N chips" counts only unresolved chips; pips cap at 8, newest first with a ring

ID	SCENARIO	EXPECTED
TT-LED-07	Mixed bet sizes queued (e.g. \$3 then \$7 chips)	Each <code>/wallet/play</code> body carries ITS chip's amount (×1,000,000) and mode — verify in the network tab

File as a launch blocker: any double deduction, any early win credit (before the line reaches the cell), or any post-session mismatch between the pill and the RGS balance.

5 Multi-Chip & Steering (TT-MCH)

ID	SCENARIO	EXPECTED
TT-MCH-01	Two winners in one column	Line converges on one, fills it, then sweeps to the other — BOTH visibly hit
TT-MCH-02	Two losers in one column	Line holds a miss lane clear of BOTH cells; neither is touched during the reveal window
TT-MCH-03	Winner + loser in one column	Line enters the winner, never grazes the loser (margin clamp)
TT-MCH-04	Any resolution	The visual outcome ALWAYS matches the paid outcome — a paid win whose cell the line missed is a launch blocker
TT-MCH-05	Chip resolves while its outcome arrives late	Fallback: it still resolves correctly by the end of the reveal window (1.6 s)
TT-MCH-06	Background the tab 30 s mid-flight, return	No burst of invisible resolutions — the sim backlog is dropped; chips resolve on-screen after return
TT-MCH-07	Camera	A resolving chip never slides off-screen; the view frames the line + nearest live column

6 Bet Picker & Currency (TT-BET)

ID	SCENARIO	EXPECTED
TT-BET-01	Quick chips (offline)	\$1 / \$2 / \$5 / \$20 snapped onto the grid; active chip highlighted; selection persists across reload
TT-BET-02	"+" menu	Full grid in a 3-column layout; top value renders MAX (tooltip = real amount); a non-quick pick fills the single custom slot, highlighted
TT-BET-03	LIVE operator grid	Picker rebuilt entirely from <code>config.betLevels</code> ; every playable amount is on-grid (no <code>ERR_VAL</code> possible via UI)
TT-BET-04	Menus exclusivity	Only one of bet menu / theme menu / settings / history open at a time; outside click and Escape close them
TT-BET-05	? <code>currency=EUR&lang=de</code>	Pill "2.000,00 €", chips "1 € ... 20 €", canvas chip labels, floats, strip, history — ALL in EUR/de format; no "\$" anywhere
TT-BET-06	<code>?currency=JPY</code>	Zero-decimal formatting (¥1,000)
TT-BET-07	Crypto/unknown code	Code-prefix fallback ("XYZ 1.50") — no crash, no NaN
TT-BET-08	Cell hover / armed preview	Shows "<mult> · win <amount>" for the CURRENT bet size (+ "tap again to confirm" when armed); hovering a live chip shows "pays <amount>"

7 Settings & Jurisdiction (TT-SET)

ID	SCENARIO	EXPECTED
TT-SET-01	Speed 1x → 2x	Chart/pace visibly doubles (sim clock only); chip approach times halve; ODDS UNCHANGED; persists (<code>taptrade.speed</code>)
TT-SET-02	Speed at 1x	Base pace is already 1.3x real time by design (see Not-A-Bug #6)
TT-SET-03	Sound off	No place/win/loss cues; persists; win/loss visuals unaffected
TT-SET-04	Sound character	Place = short tick · loss = low thud · win = 2/3/4-note arpeggio for normal/big(≥5x)/epic(≥20x)
TT-SET-05	Tap-to-confirm ON	First tap/click previews (outlined cell + offer text), second tap on the SAME cell places; tapping elsewhere re-arms; works with a mouse too
TT-SET-06	Tap-to-confirm OFF	Single tap places immediately
TT-SET-07	Defaults	Touch device → confirm ON; mouse → OFF (fresh storage)
TT-SET-08	<code>jurisdiction.disabledTurbo</code> (mock: edit mockRgs JURISDICTION)	1.5x/2x hidden; a persisted fast speed resets to 1x

ID	SCENARIO	EXPECTED
TT-SET-09	<code>jurisdiction.socialCasino</code>	Bet row label reads "Play"

8 History & Replay (TT-HIS)

ID	SCENARIO	EXPECTED
TT-HIS-01	Resolve a chip, open the panel (clock button or strip pips)	Row: JPEG thumbnail (the reveal state visible — gold fill or red flash), $\pm$ amount colored, "<mult> · <stake> stake", clock time
TT-HIS-02	Landing shot content	Crop centered on the cell, captured ~0.5 s AFTER the reveal so the outcome state shows
TT-HIS-03	Two chips in one column resolve in the same tick	TWO distinct history rows (unique keys — the <code>each_key_duplicate</code> regression)
TT-HIS-04	Modal replay	Faithful mini-chart: real cell grid, game aspect ratio, line animates ~1.4 s and ends where the landing shot shows; cell fills with the outcome + "+amount"/"MISS"
TT-HIS-05	Cap & persistence	Max 24 entries (oldest drop); history clears on reload (in-memory by design)
TT-HIS-06	25+ resolutions	No unbounded memory growth; panel scrolls with the slim themed scrollbar

9 Splash, Errors & Resume (TT-SPL / TT-ERR / TT-RES)

ID	SCENARIO	EXPECTED
TT-SPL-01	Fresh load	Loader webp loops once (~2.2 s) on the dark-teal ground → loader fades → ground eases to black → sting video fades in (sound if permitted, else muted, else skipped)
TT-SPL-02	Skip paths	Escape always dismisses; tap dismisses once the sting is playing; video end/error dismisses; hard failsafe at 20 s — the player can NEVER be trapped
TT-SPL-03	Under the splash	The game is already running (no dead time after dismissal)
TT-ERR-01	Dismissible errors (e.g. <code>ERR_IPB</code> , <code>ERR_GEN</code>)	Themed modal, friendly text, "Dismiss" button; game continues
TT-ERR-02	Fatal codes <code>ERR_IS</code> / <code>ERR_ATE</code>	Modal is NOT dismissible — single accent "Reload" button; reload re-authenticates
TT-ERR-03	Error copy	Codes map to friendly text (session expired / insufficient balance / invalid bet / limit / location / maintenance ...) — never a raw code alone
TT-ERR-05	Production build, paramless launch	Blocking overlay "This game must be launched from the casino." — no play-money fallback, no bets possible

ID	SCENARIO	EXPECTED
TT-ERR-06	Production build + <code>?demo=1</code>	Play-money works exactly like dev
TT-ERR-07	Production build console	<code>window.__tt</code> / <code>__ttHistory</code> are UNDEFINED
TT-RES-01	LIVE: refresh mid-round (chip in flight), relaunch	Round settles silently on authenticate; toast "Last round settled: + <amount>" (win) or "Last round settled"; balance correct; next bets work (no "active round" lockout)
TT-RES-02	After resume	No ghost chip is painted (the cell position was never persisted — by design)

10 Mobile & Accessibility (TT-MOB)

ID	SCENARIO (≤ 700 PX WIDTH)	EXPECTED
TT-MOB-01	Bottom bar	Bet row spans full width, "Bet" label hidden, buttons ≥ 44 px, honors the safe-area inset
TT-MOB-02	Zoom rail	Docked vertically at mid-left
TT-MOB-03	Header	Compact brand (88 px) + pill at left:110; session strip wraps below, full width, rounded-14
TT-MOB-04	Menus/panels	Stretch edge-to-edge (12 px margins), capped at 48 vh, above the bottom bar
TT-MOB-05	Title badge	Hidden
TT-MOB-06	Pinch zoom	Two-finger pinch zooms; releasing does NOT place a ghost chip (400 ms tap guard)
TT-MOB-07	Toasts / hint	Toasts stretch and wrap; hint wraps at 82 vw
TT-MOB-08	<code>prefers-reduced-motion</code>	No chip shake/pulse/vignette, no menu/switch/splash/hint transitions, replay renders its final frame instantly — chips still resolve correctly
TT-MOB-09	Keyboard / ARIA	Menus are <code>role="menu"</code> with <code>menuitemradio</code> + <code>aria-checked</code> ; switches are <code>role="switch"</code> ; focus-visible outlines on all controls; balance pill is <code>aria-live="polite"</code>

11 Math Spot-Checks & Reference

11.1 Network-level checks (mock or LIVE)

- Every `/wallet/play` body: `mode = "call_<cents>"` matching the tapped cell's displayed multiplier ×100; `amount = stake × 1,000,000`.
- Win round: `round.active:true` in the play response, then exactly one `/wallet/end-round`. Loss round: `round.active:false`, NO end-round call.
- Book events in `round.state`: win = `cellCall → wincap → finalWin` (wincap on EVERY win — see Not-A-Bug #1); loss = `cellCall → finalWin(amount:0)`.
- Scales: event amounts are bet-cents (450 = 4.5×); `round.payoutMultiplier` is PLAIN (4.5); API money ×1,000,000.

11.2 Displayed multipliers

Every multiplier shown on a cell must be one of the 28 ladder values below — nothing else may ever be offered. (Cell placement of a given value varies with the volatility model; the *values* may not.)

MODE	MULTIPLIER	WIN CHANCE (A/B)	WIN %	RTP	BOOKS
call_140	1.4×	11/16	68.75%	96.25%	16

MODE	MULTIPLIER	WIN CHANCE (A/B)	WIN %	RTP	BOOKS
call_150	1.5×	9/14	64.29%	96.43%	14
call_160	1.6×	3/5	60.00%	96.00%	5
call_180	1.8×	8/15	53.33%	96.00%	15
call_200	2×	12/25	48.00%	96.00%	25
call_220	2.2×	7/16	43.75%	96.25%	16
call_250	2.5×	5/13	38.46%	96.15%	13
call_280	2.8×	10/29	34.48%	96.55%	29
call_320	3.2×	3/10	30.00%	96.00%	10
call_360	3.6×	4/15	26.67%	96.00%	15
call_400	4×	6/25	24.00%	96.00%	25
call_450	4.5×	3/14	21.43%	96.43%	14
call_500	5×	5/26	19.23%	96.15%	26
call_600	6×	4/25	16.00%	96.00%	25
call_700	7×	4/29	13.79%	96.55%	29
call_800	8×	3/25	12.00%	96.00%	25
call_900	9×	3/28	10.71%	96.43%	28
call_1000	10×	5/52	9.62%	96.15%	52
call_1200	12×	2/25	8.00%	96.00%	25
call_1500	15×	5/78	6.41%	96.15%	78
call_2000	20×	4/83	4.82%	96.39%	83
call_2500	25×	1/26	3.85%	96.15%	26
call_3000	30×	4/125	3.20%	96.00%	125
call_4000	40×	2/83	2.41%	96.39%	83
call_5000	50×	1/52	1.92%	96.15%	52
call_6500	65×	2/135	1.48%	96.30%	135
call_8000	80×	1/83	1.20%	96.39%	83
call_10000	100×	1/104	0.96%	96.15%	104

Band 96.00–96.55% · spread 0.5517% · 1,176 books. Long-run empirical win rates per mode should trend toward the table (LOCAL uses the same published weights).

12 Not-A-Bug List (do not file)

#	OBSERVATION	WHY IT'S INTENTIONAL
1	A <code>wincap</code> event on every winning round, even a 1.4x win	Per-mode <code>max_win</code> equals the mode's own multiplier (math-side design); the client no-ops it
2	Only 2 chips per column, even with a gap between cells	Static cap — a 3rd chip can demand an undrawable win-lose-win pattern; a dynamic cap would leak drawn outcomes
3	Balance doesn't change when a chip explodes	The stake left at placement; "Missed" is an outcome marker, not a debit
4	Win credit waits for the line to reach the cell	Reveal-time crediting is the masking design — an early credit would spoil the outcome
5	The same multiplier appears in different places over time	Cell pricing follows a live volatility model; only the VALUES are fixed (the ladder)
6	Game feels faster than real time at speed 1x	Base sim pace is 1.3x by design; the setting multiplies it
7	History clears on reload	Landing-shot JPEGs are too large for <code>localStorage</code> — in-memory only
8	Balance pill isn't clickable (demo resets on click in LOCAL)	The app's wallet is mock/RGS-owned; no reset affordance on purpose
9	No "Try again" button on errors (demo has one)	The app compiles its odds in — there is no retryable load; Reload covers recovery
10	Resume shows no chip on the board	Books are position-neutral; the tapped cell is never persisted
11	Prices/line movement don't match any real market	The chart is presentation over pre-generated outcomes (certified replay); prices are theatre